Communication Protocol

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What is CAN ?

- **Controller Area Network** (CAN) was initially created by German automotive system supplier Robert Bosch in the 1983 for automotive applications.
- CAN became ISO standard 11898 and SAE J1939.
- CAN used especially for reliable data exchange between electronic control units (ECUs) in the automobile.
- CAN controller chip were introduced by Intel and after that Philips.

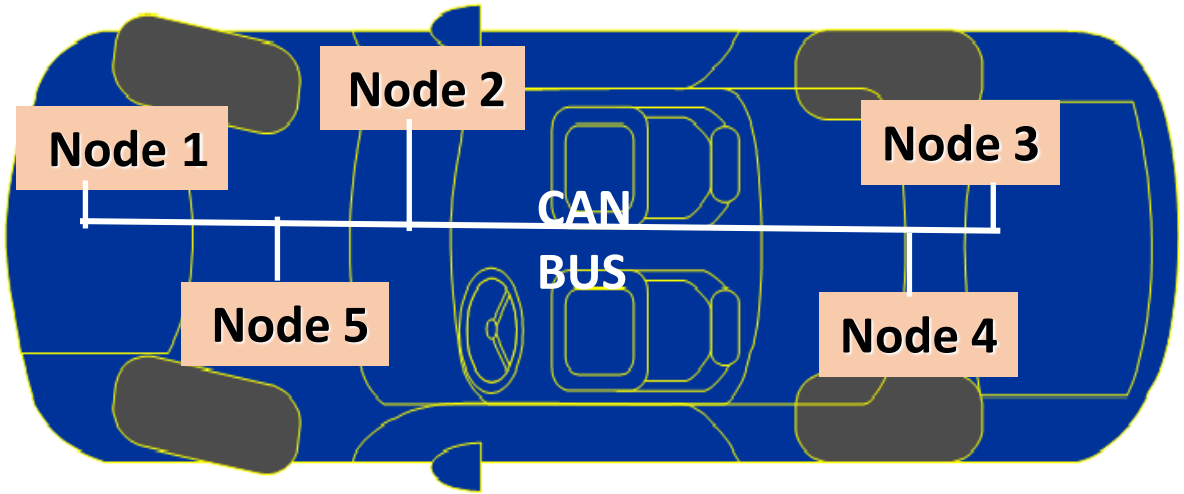


Source: https://www.allaboutcircuits.com/technical-articles/introduction-to-can-controller-area-network/

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Continued..

- All nodes attach to common connection.
- All nodes on the CAN network must operate at the same nominal bit rate otherwise errors will occur on receiving side.
- **CAN** is a multi-master serial **bus** standard for connecting Electronic Control Units also known as **nodes**.



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CAN Features

- CAN is half-duplex high speed serial communication protocol.
- CAN is a multi-master, multicast protocol.
- CAN is fast, robust communication.
- The priority over bus is decided by the message not the node.
- It is Message based, not address based
  - Messages are NOT transmitted from one node to another node based on addresses.
  - New nodes can be added to a system without having to update all of the nodes with addressing information

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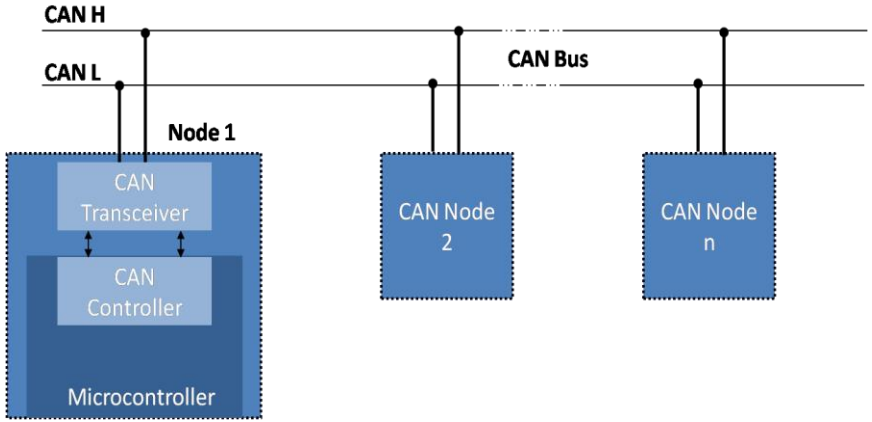
Role of CAN in Embedded system

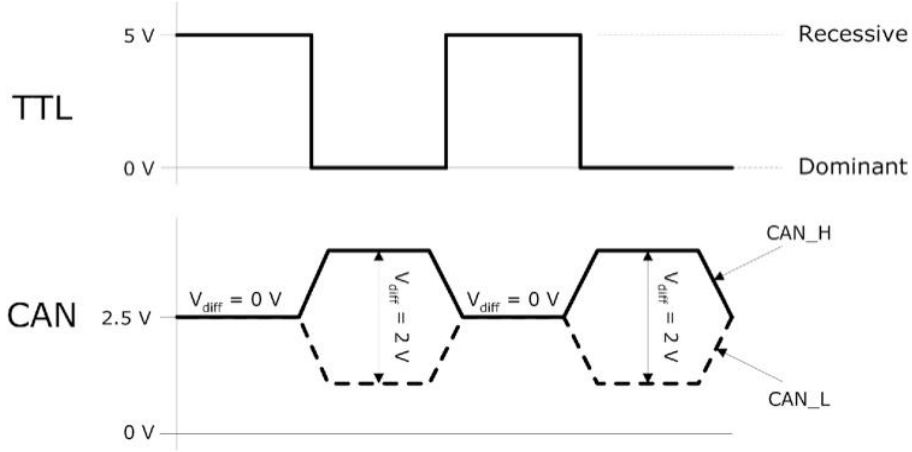
- A Controller Area Network is a robust vehicle bus standard and is mean to be designed to allow microcontrollers and devices to communicate with each other's applications without a host controller.
- CAN High Speed
  - ISO11898-2
  - Supports up to 1Mbps bus speed
- CAN Low Speed
  - ISO11898-3
  - Supports up to 125kbps bus speed
- CAN bus uses two differential signals namely CANH(High) and CANL(Low)

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Why CAN Signal used ?

- CAN needs 2 lines, CAN_L, CAN_H.
- It is a differential line, where the signal is represented by a differences between CAN_L and CAN_H. If the difference is positive and larger than a certain minimum voltage it is called ‘Dominant’ and if the difference is zero it is called ‘Recessive.’



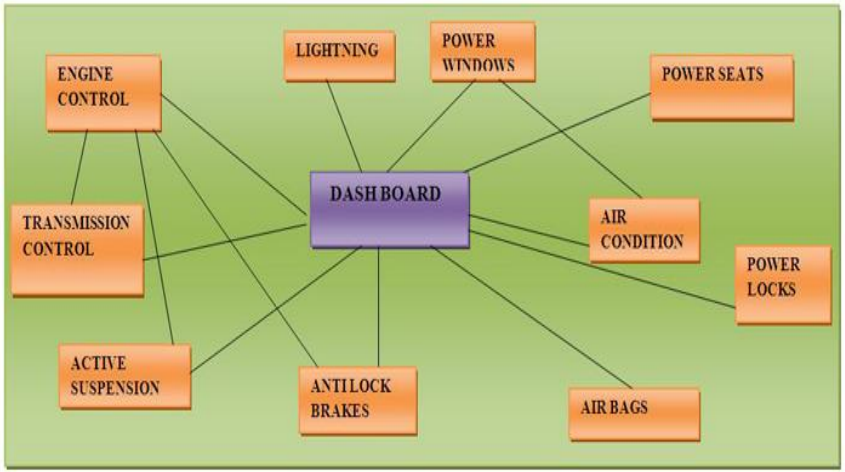


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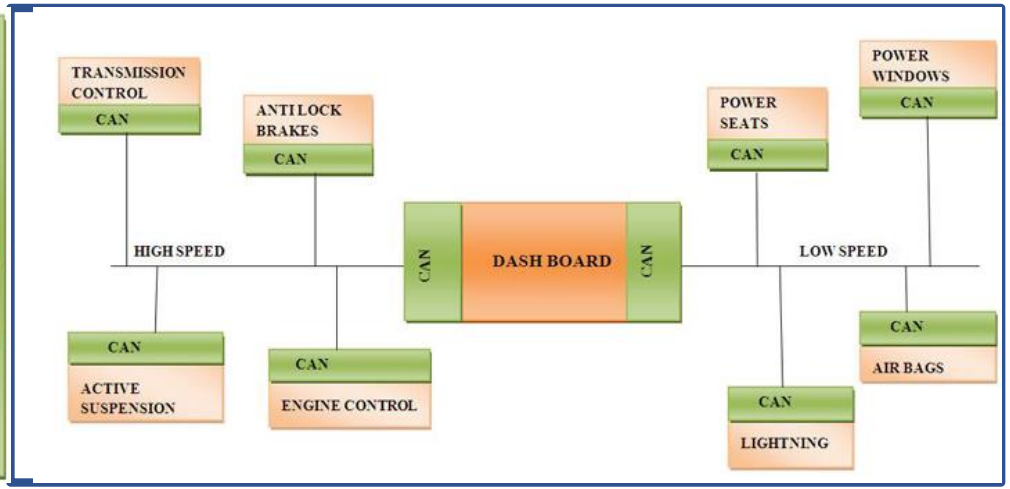
Why CAN ?

- Robust in noisy environment
- Priority signal setting
- Cost effective

Before CAN



After CAN

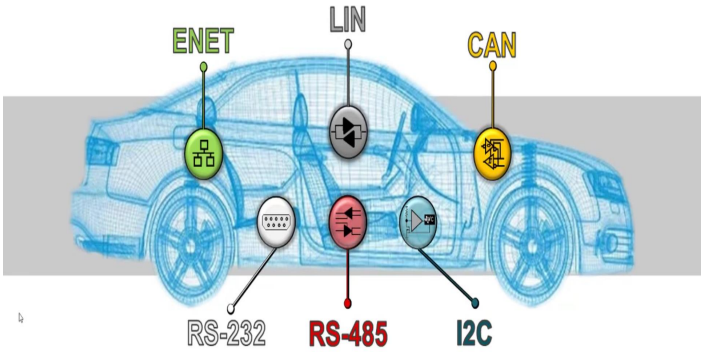


Source: https://www.engineersgarage.com/article_page/can-protocol-understanding-the-controller-area-network-protocol/

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Comparison of CAN protocol with other protocol

Protocol	Transfer Type	Transfer Rate (bits/sec)	Max.Length (m)	No. of Nodes
RS232	Point to Point	Up to 20K	15	1
RS485	Address	32K 10M	1000 4000	32
I2C	Address	Standard(100K)	1	128
CAN	Network	1M	40	-
LIN	Network	40K	40	-



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Bus Length Vs Signaling Rate

- Data rate dependent on bus length.

Signaling rate in Kbits/s	Bus length in Meters
1000	40
500	100
250	200
125	500

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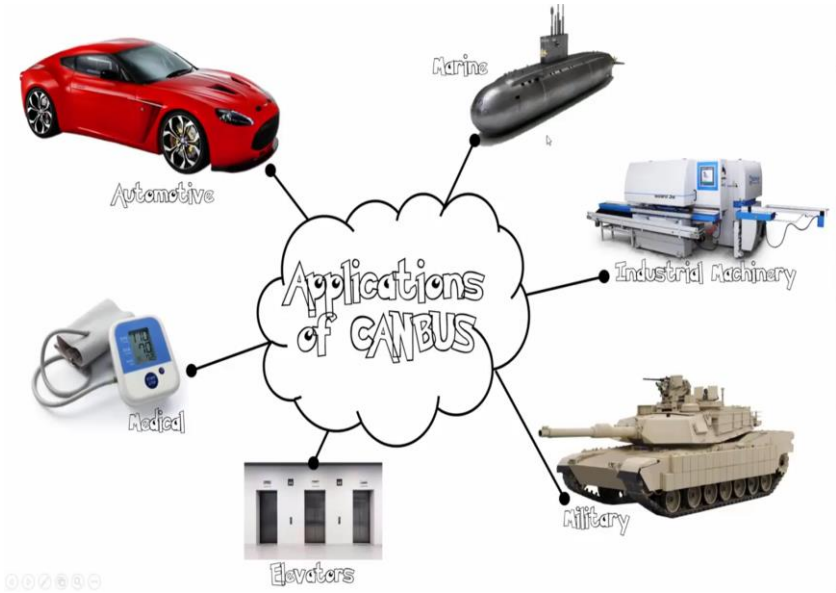
Benefits of CAN over other protocol

Advantages	
Low-Cost, Lightweight Network	•CAN provides an inexpensive, durable network that helps multiple CAN devices communicate with one another. •As CAN serial bus uses two wires, it offers good price/performance ratio. Also, driven by high volume production of low cost protocol devices, they are relatively cheap.
Speed	•CAN supports data rate of 1 Mbps at 40m bus length.
Flexibility	• CAN Nodes can be easily connected / disconnected.
Robustness	•Durability and reliability are key areas of concern when choosing a communication protocol for deployment. •The system is robust towards electric disturbances and electromagnetic interference - ideal for safety critical applications (e.g. vehicles)

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Application area of CAN

- Automotive (Passenger, Truck, Buses)
- Electronic equipment for aviation and navigation .
- Industrial automation
- Medical instruments and equipments
- Building automation



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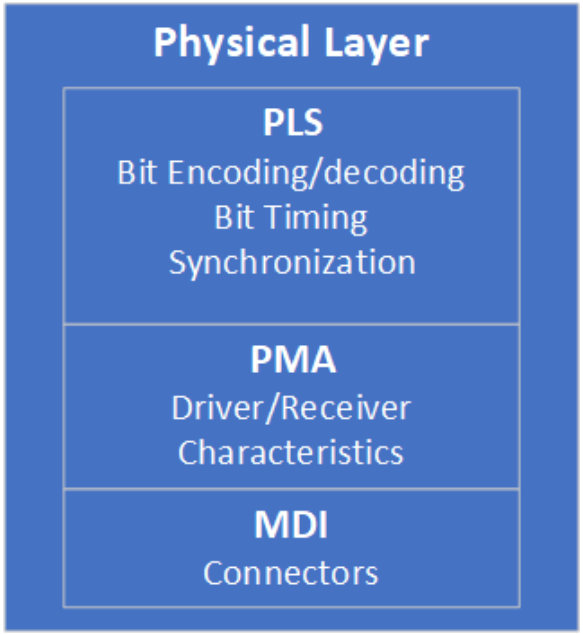
History of CAN in short

- **Pre CAN:** Car ECUs relied on complex point-to-point wiring
- **1986:** Bosch developed the CAN protocol as a solution
- **1991:** Bosch published CAN 2.0 (CAN 2.0A: 11 bit, 2.0B: 29 bit)
- **1993:** CAN is adopted as international standard (ISO 11898)
- **2003:** ISO 11898 becomes a standard series
- **2012:** Bosch released the CAN FD 1.0 (flexible data rate)
- **2015:** The CAN FD protocol is standardized (ISO 11898-1)

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CAN Physical Layer

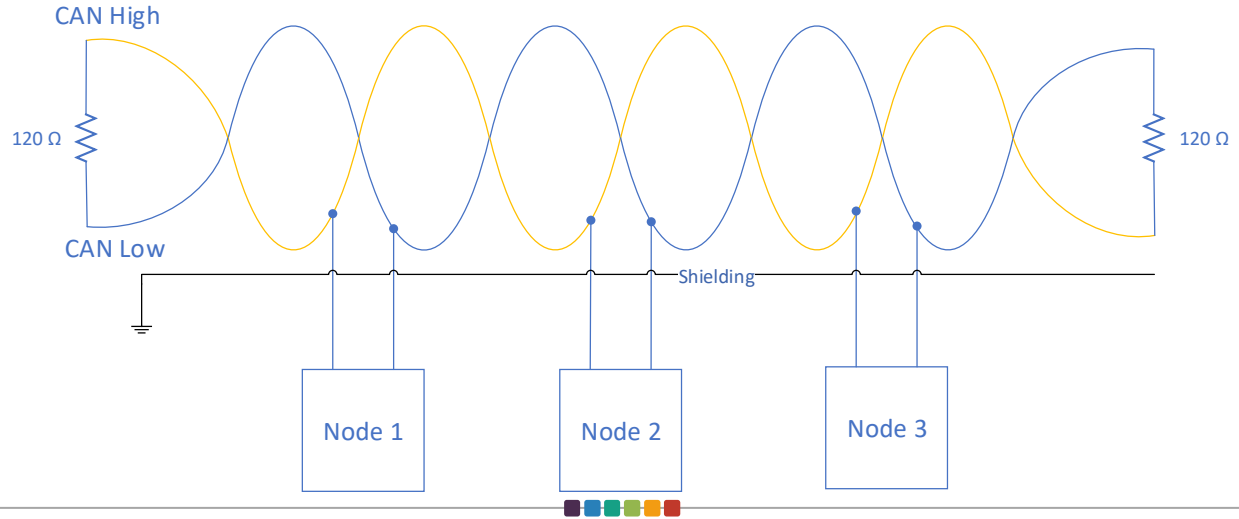
- PLS – Physical Signaling
- PMA – Physical Medium Attachment
  - Specifies transceiver characteristics
- MDI – Medium Dependent Interface
  - Mechanical and Electrical interface between physical medium and MAU (Medium Access Unit)



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CAN Bus connections

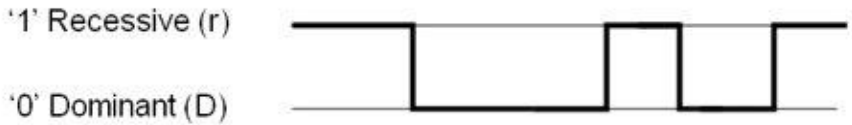
- Two wires used for communication are named as
  - CAN High (CAN H)
  - CAN Low (CAN L)
- Bus is terminated with 120 Ω resistor on both ends



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CAN Bus properties

- Uses wired-and logic



Two logic states on the CAN bus

Node 1	Node 2	Node 3	Bus
D	D	D	D
D	D	r	D
D	r	D	D
D	r	r	D
r	D	D	D
r	D	r	D
r	r	D	D
r	r	r	r

Wired-AND Function

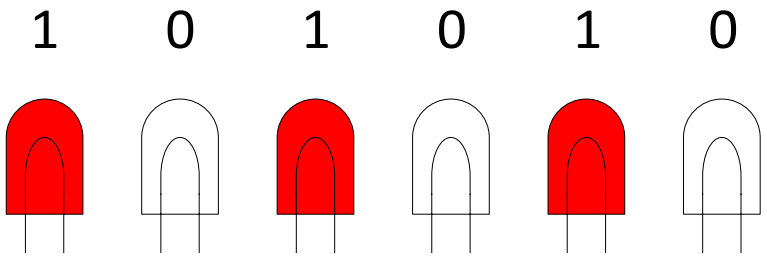
Bus in dominant state

Bus in recessive state or idle

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Bit Encoding/Decoding

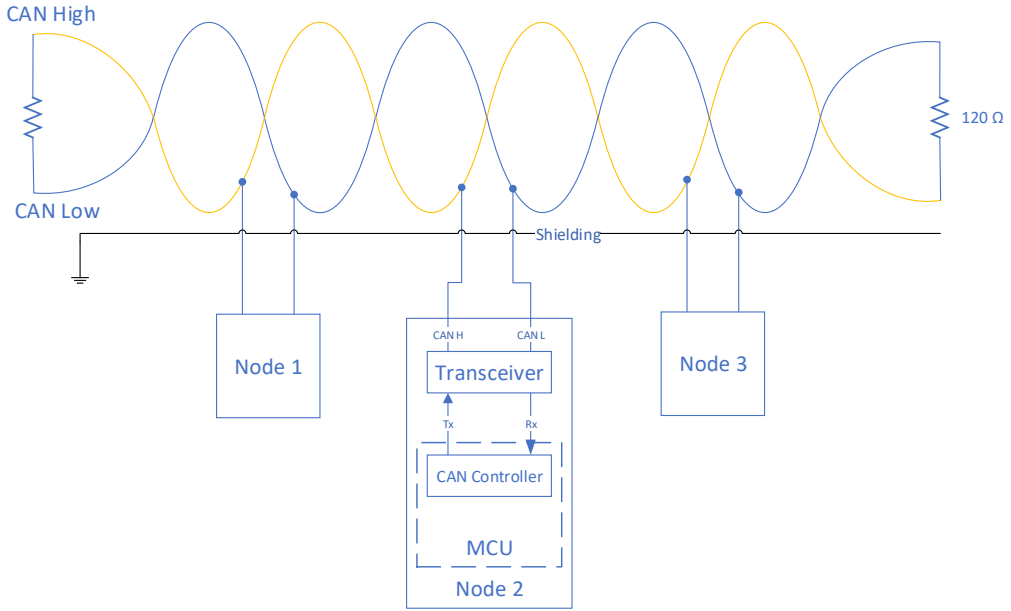
- Nominal Bit Rate (BR) - Number of bits transmitted per second
- Nominal Bit Time – Time required for sending one bit (1/BR)
- Encoding method defines how each bit is transferred to the PMA sublayer
- NRZ unipolar encoding is used for CAN



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CAN Node Components

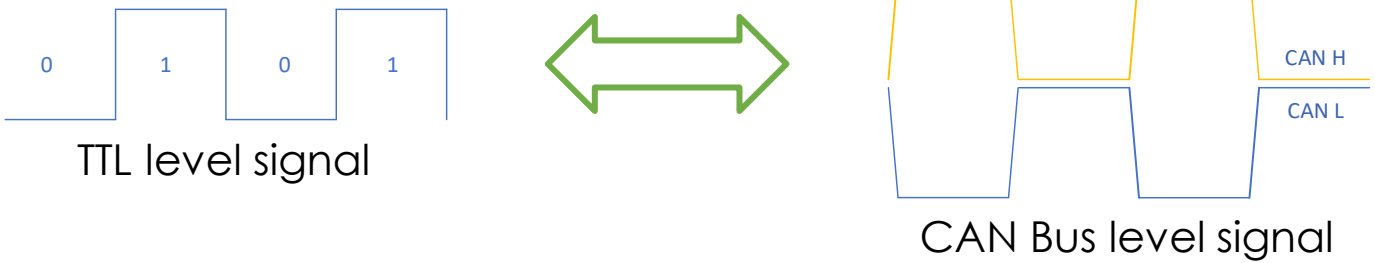
- Each CAN node contains
  1. Micro-controller
  2. CAN controller
  3. CAN Transceiver



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Transceiver Advantages

- Saves crucial PCB space
  - Protection circuit
  - Signal level conversion circuit
- MCU is protected from high voltage bus faults
- Provides ESD protection
- Handles conversion of TTL level binary data to CAN bus waveforms and vice versa



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Standard Bit rates

- ISO 11898-2 specifies properties of High-Speed CAN
  - Speeds up to 1 Mbps
- ISO 11898-3 specifies properties of Low Speed CAN
  - Speeds from 40 Kbps to 125 Kbps
- In general, with 1 Mbps speed, 40m cable length can be supported
- Generally, following standard baud rates are used for CAN communication
  - 125 Kbps
  - 250 Kbps
  - 500 Kbps
  - 1000 Kbps
- Passenger vehicles use 500 Kbps for diagnostics ports
- Commercial vehicles use 250 Kbps for diagnostics ports

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CAN Bus Diagnostics

- Resistance between CANH and CANL wire should be around 60 Ω
- Voltages of the CAN Bus when active, should be between allowed range of recessive level
- Voltage measured between CANH and CANL should be 0V
- If all of the physical connections are correct, then Baud rate should be verified

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LIN Protocol

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LIN Basics

- Local Interconnect Network
- Serial network protocol used for communication between components in vehicles.
- The need for a cheap serial network arose as the technologies and the facilities implemented in the car grew, while the CAN bus was too expensive to implement for every component in the car.



What is LIN?

- The core of LIN is a serial network protocol controller and a physical layer (PHY) transceiver (TRX).
- The TRX transmitter converts the logic level LIN protocol data stream on the TXD input into V_{BAT} level LIN bus signal using a current – limited wave – shaping driver which reduces electromagnetic emissions (EME).
- The TRX receiver converts the high voltage data stream from the LIN bus to logic level signals that are sent to the microprocessor through the RxD pin.
- The network protocol controller provides the synchronization, logic, error detection, and other features that are usually associated with LIN.
- The network relies on a single master communicating with up to 16 slave devices.

LIN Concept

- Technical Solution

  a) Low cost single-wire implementation (enhanced ISO 9141)

  b) Speed up to 20Kbit/s (limited for EMI-reasons)

  c) Single Master / Multiple Slave Concept

  → No arbitration necessary

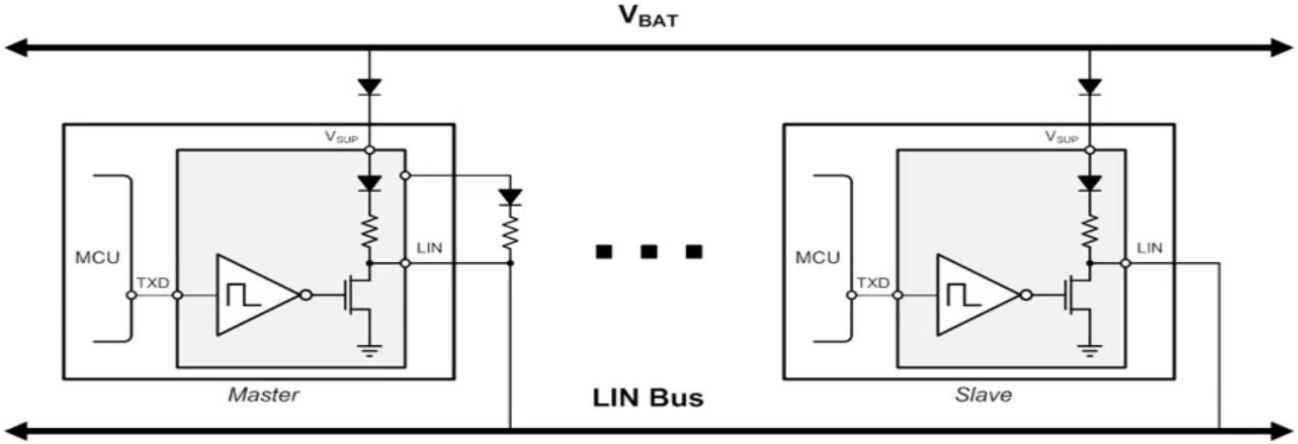
  d) Low cost silicon implementation based on common UART/SCI interface hardware

  → Almost any Microcontroller has necessary hardware on chip

  e) Self synchronization without crystal or ceramics resonator in the slave nodes

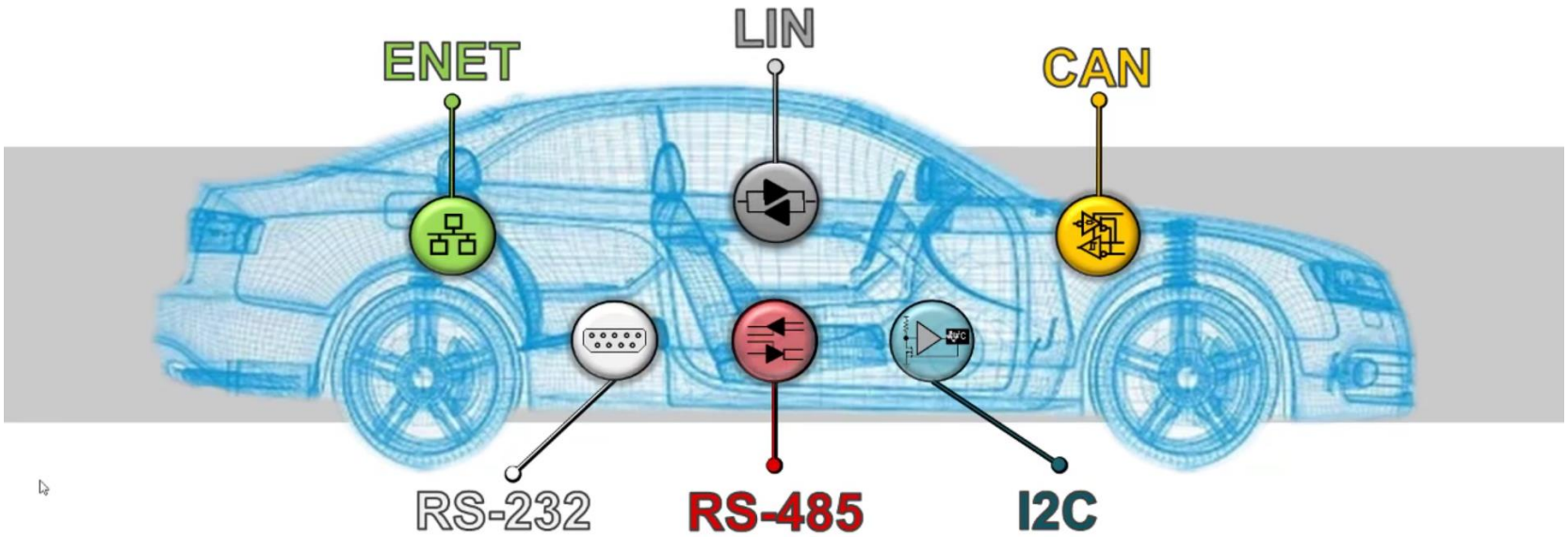
  → Significant cost reduction of hardware platform

Local interconnect network

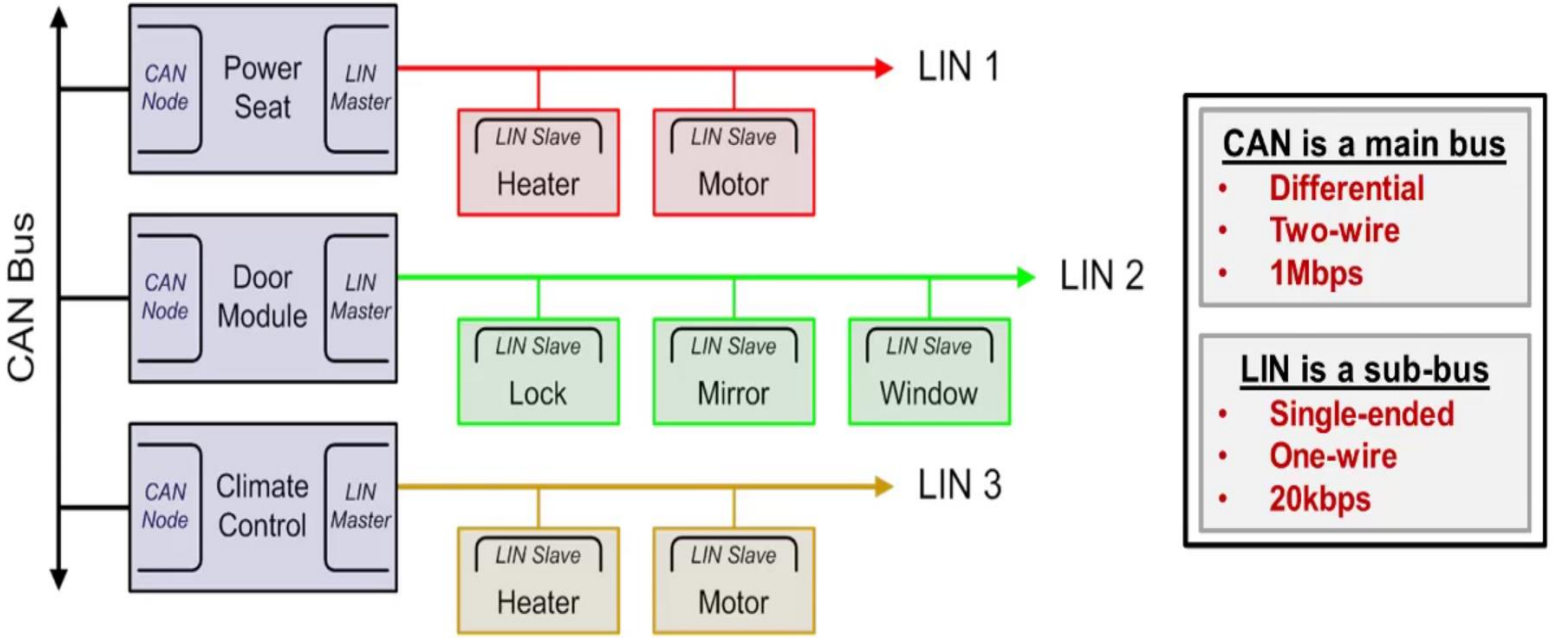


- LIN is a broadcasting, serial, one-wire interface, typically implemented as a sub-bus of a CAN network.
- Allows automotive manufacturers to reduce cost by offloading low-speed (<20kbps), non-safety critical functions from a two-wire CAN bus to one-wire bus.
- One master coordinates communication between up to 16 slaves.

LIN Interface for Automotive Application

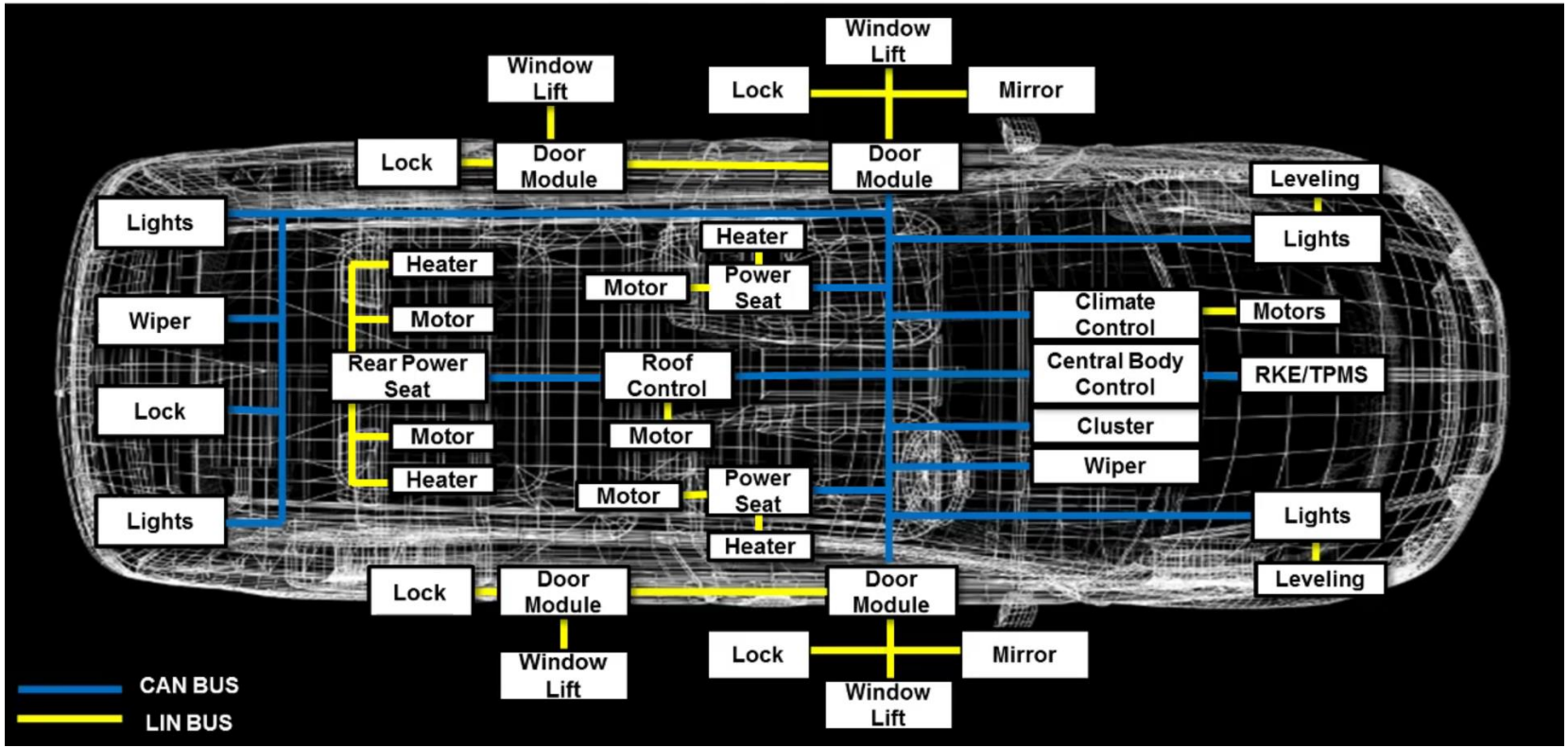


CAN & LIN in Automotive Application



Automotive Body Network Architecture

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LIN Consortium

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- Consortium formed in 1998.

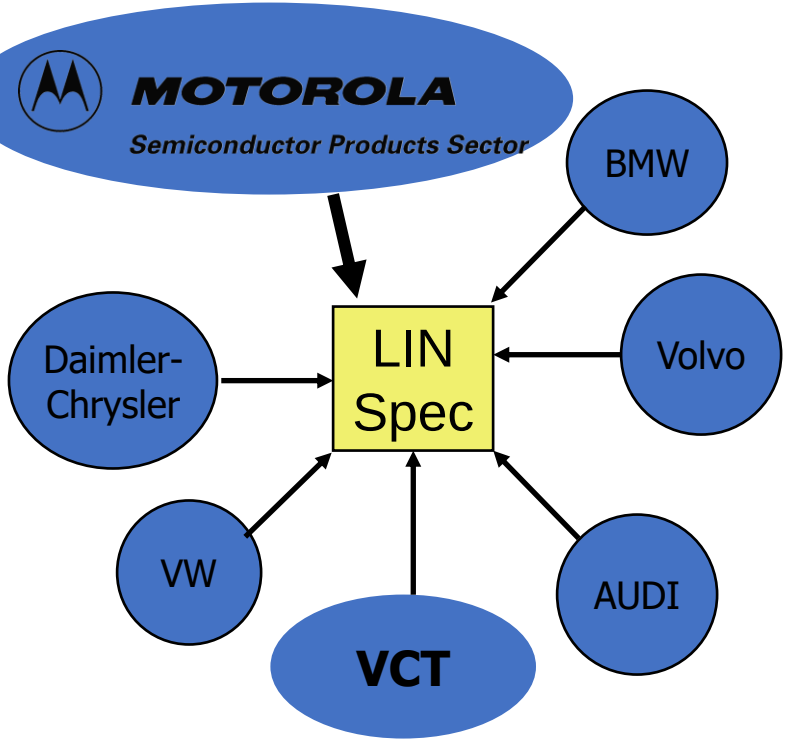
Five Car manufacturers
ONE Semiconductor Supplier (Motorola)
One tool Supplier (VCT)

- Specification finalized on 02/02/00

Official Launch at SAE March ‘00
Open Specification.

- Motorola Ready to support LIN with extensive device families and new parts already in the discussion/ spec finalization loop.

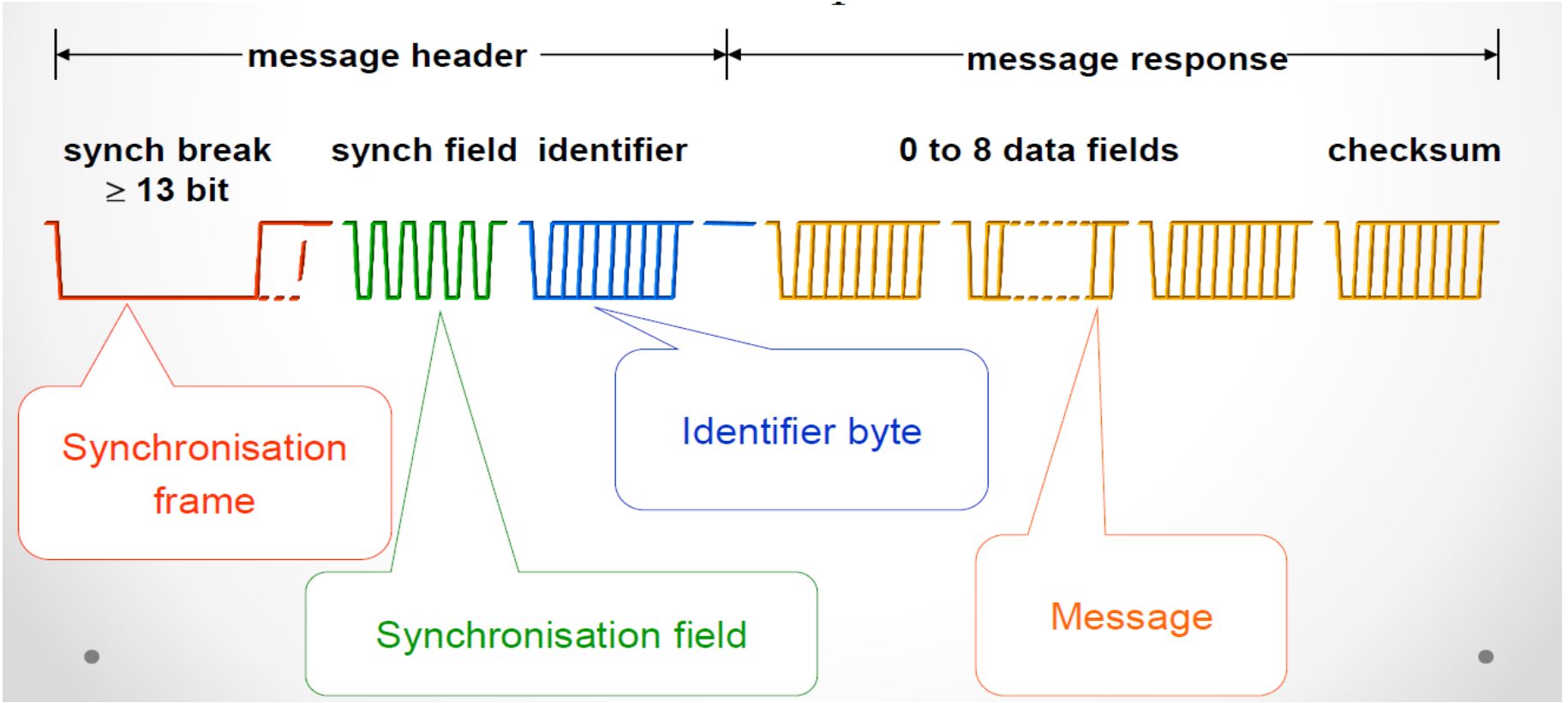
First dedicated LIN part available Q3 ‘00



Revision History

LIN Versions	Date	Remark
LIN 1.0	1999-07-01	Initial Version of the LIN Specification
LIN 1.1	2000-03-06	-
LIN 1.2	2000-11-17	-
LIN 1.3	2002-12-13	-
LIN 2.0	2003-09-16	Major Revision Step
LIN 2.1	2006-11-24	Clarifications, configuration modified, transport layer enhanced and diagnostics added.
LIN 2.2	2010-12-31	According to LIN 2.1 Errata Softened bit sampling specification
LIN 2.2A	2010-12-31	Corrected wakeup signal definition in chapter 2.6.2

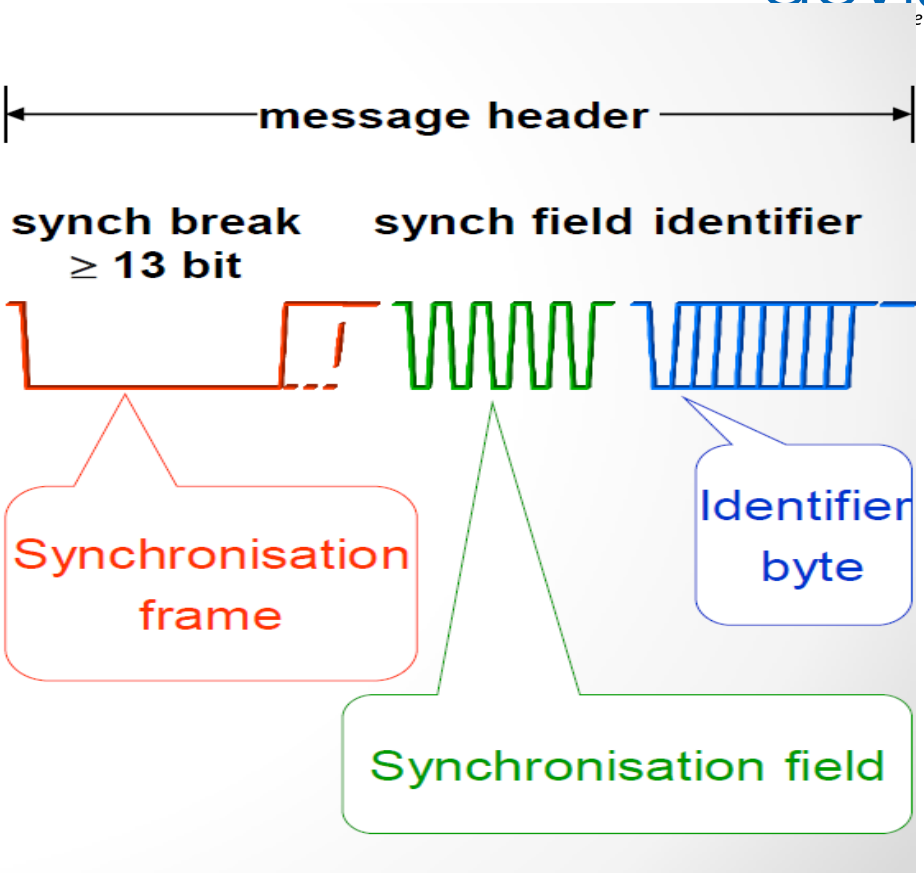
LIN Message Frame (1)



LIN Message Frame (2)

Master Task:-

- Control over the whole Bus.
- Controls which message at what time is to be transferred over the bus.
- Send Header: Sync Break, Sync Byte ,ID-Field.



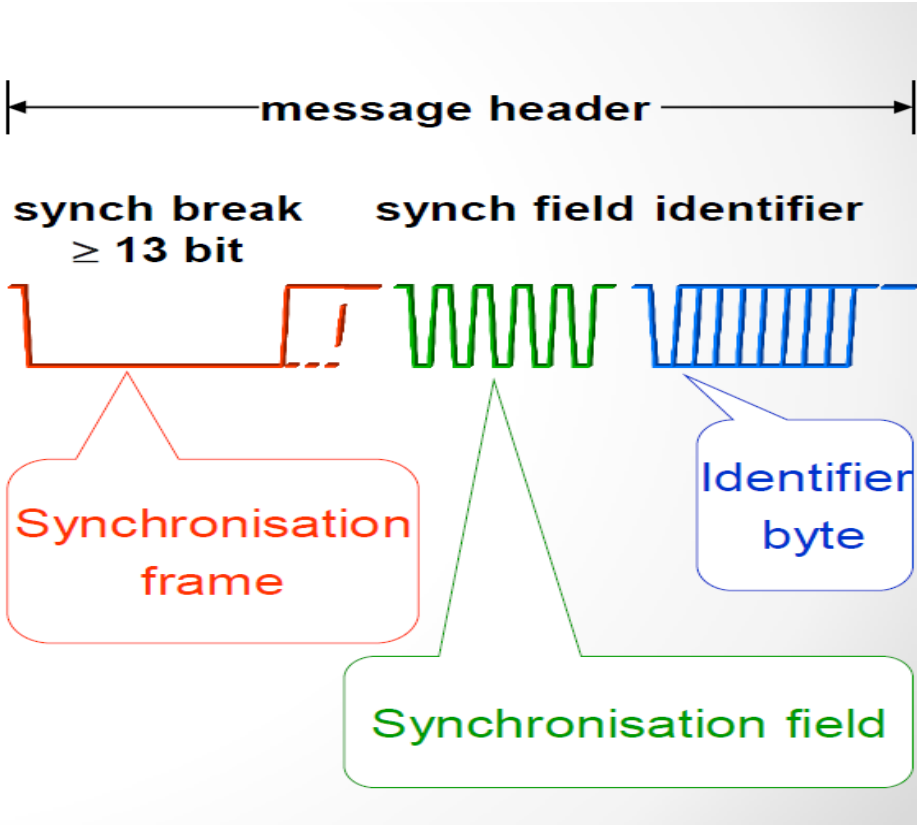
LIN Message Frame (3)

Master Task:-

- Error handling.

1. monitors Data Bytes and Check Byte, and evaluates them on consistence

- Receives Wakeup Break from slave nodes when the bus is inactive.
- Defines the transmission speed.
- Switching slave nodes to sleep/wake up mode.

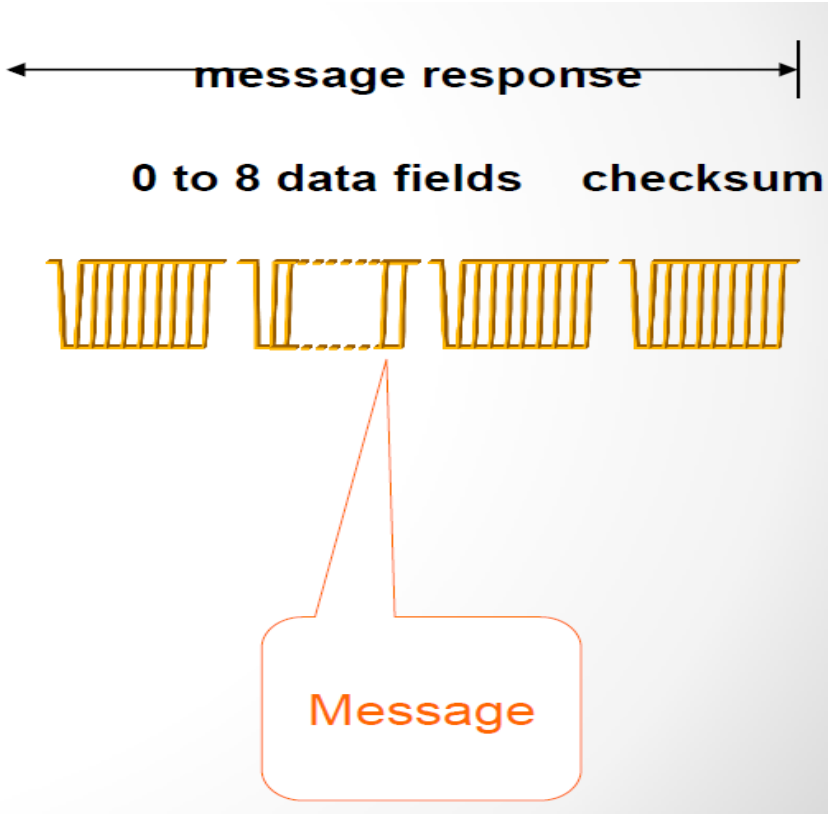


LIN Message Frame (4)

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Slave Task:-

- One of 2-16 Members on the Bus.
- Receives or transmits Data when appropriate ID is sent.
- Slave snoops for ID.



LIN Message Frame (5)

Slave Task:-

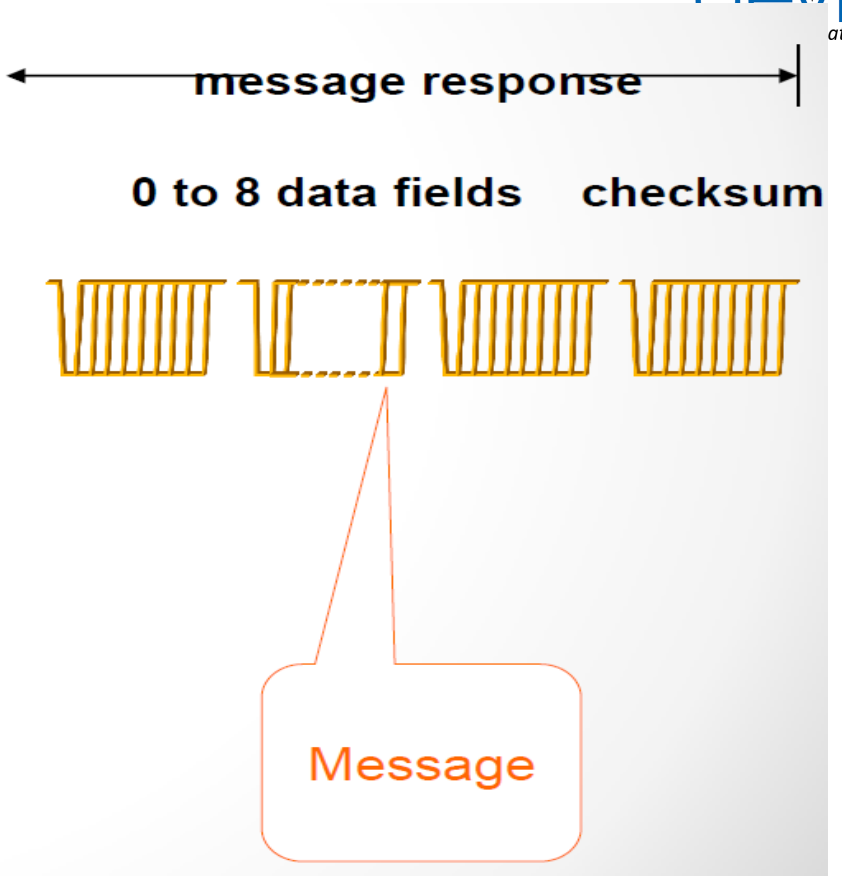
- According to ID, slave determines:

1. receive data, transmit data ,do nothing.

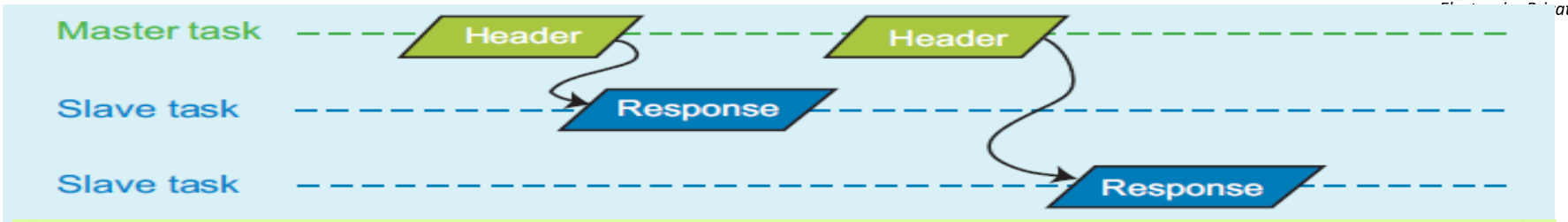
- When transmitting :

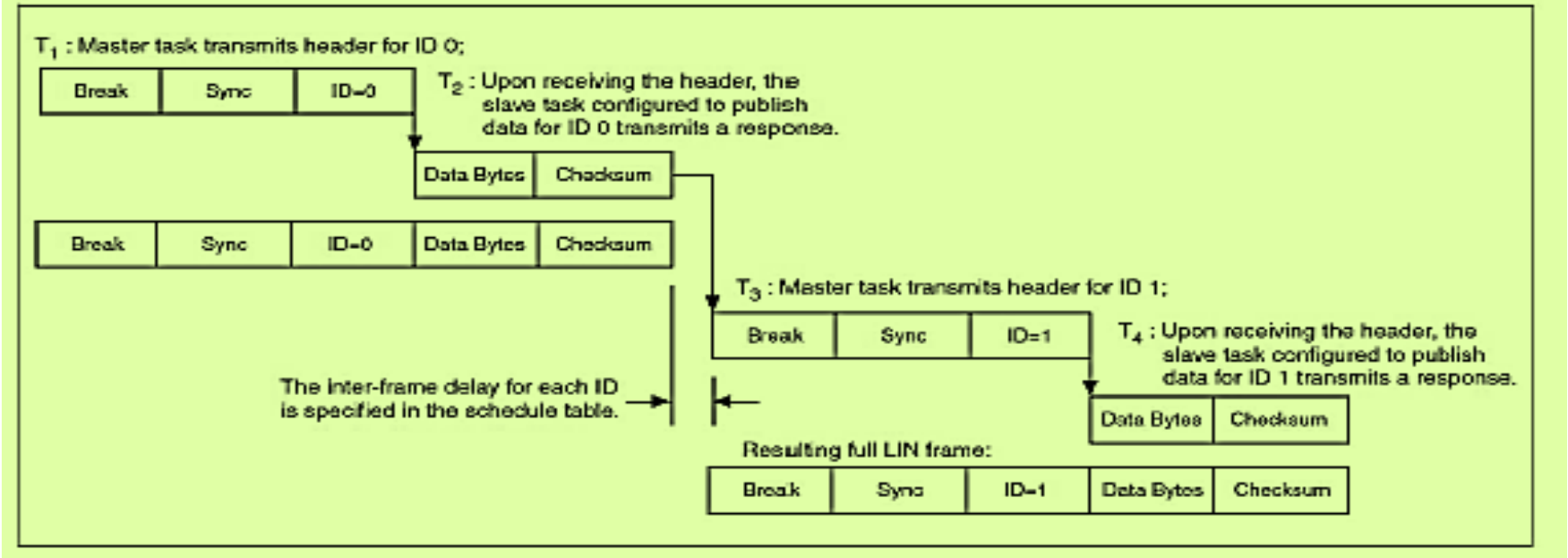
1. sends 1, 2, 4, or 8 Data Bytes + Check-Byte

- The node serving as a master can be slave.

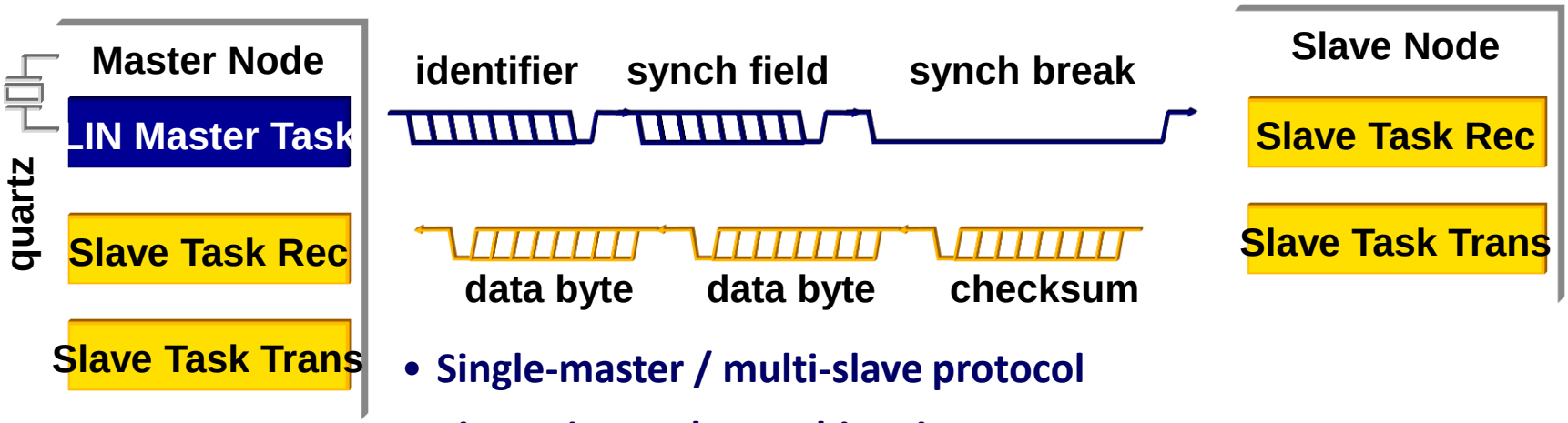


Master and Slave Communication



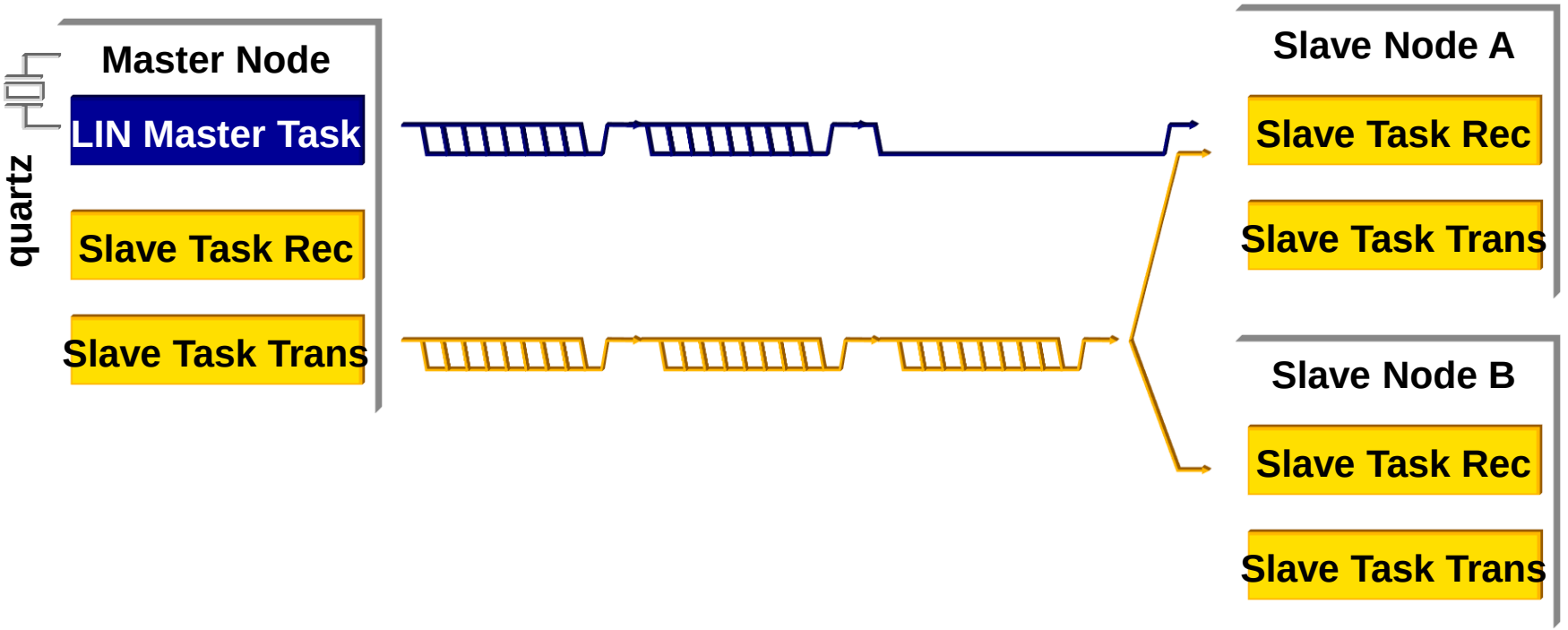


LIN Comm. – 1. Data from Slave to Master

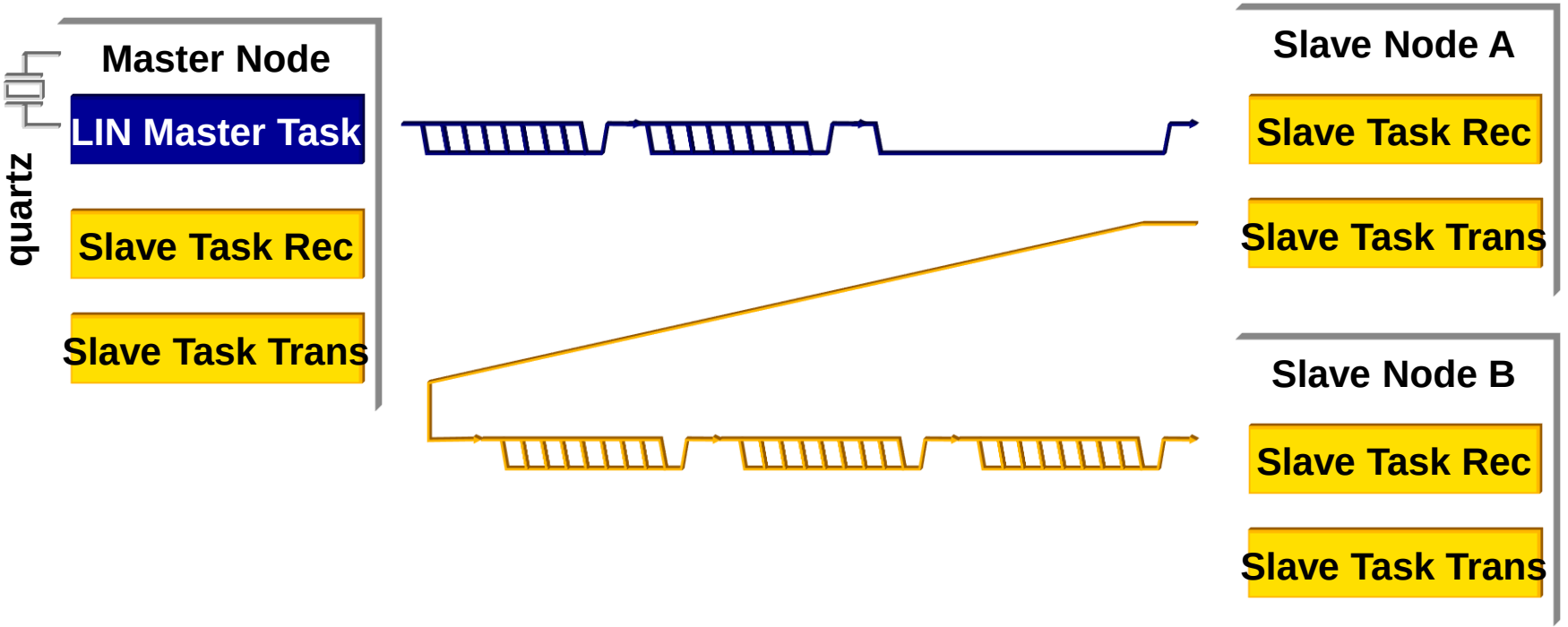


- Single-master / multi-slave protocol
- Time triggered, no arbitration
- Identifier denotes message content, not physical address
- Multicast messages
- Baud rate synchronization through protocol
- Power saving sleep mode

LIN Comm. – 2. Data from Master to Slave (s)



LIN Comm. – 3. Data from Slave to Slave



Ethernet

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Introduction Networking

The past few decades have witnessed an exponential increase in the number and sophistication of electronic systems in vehicles.

Today, the cost of electronics in luxury vehicles can amount to more than 40 percent of the total manufacturing cost



Complex Network System





•The resulting demands on power and design have led to innovations in electronic networks for automobiles

•Just as LANs connect computers, control networks connect a vehicle's electronic equipment. These networks facilitate the sharing of information and resources among the distributed applications.

•In the past, wiring was the standard means of connecting one element to another.

Impact on increase in Electronics

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- Added wiring increased vehicle weight, and weakened performance made adherence to reliability standards difficult

- Complex wiring harnesses took up large amounts of vehicle volume, limiting expanded functionality and eventually became the single most expensive and complicated component in vehicle electrical systems

- Today's control and communications networks, based on serial protocols, counter the problems of large amounts of discrete wiring



In-Vehicle Network

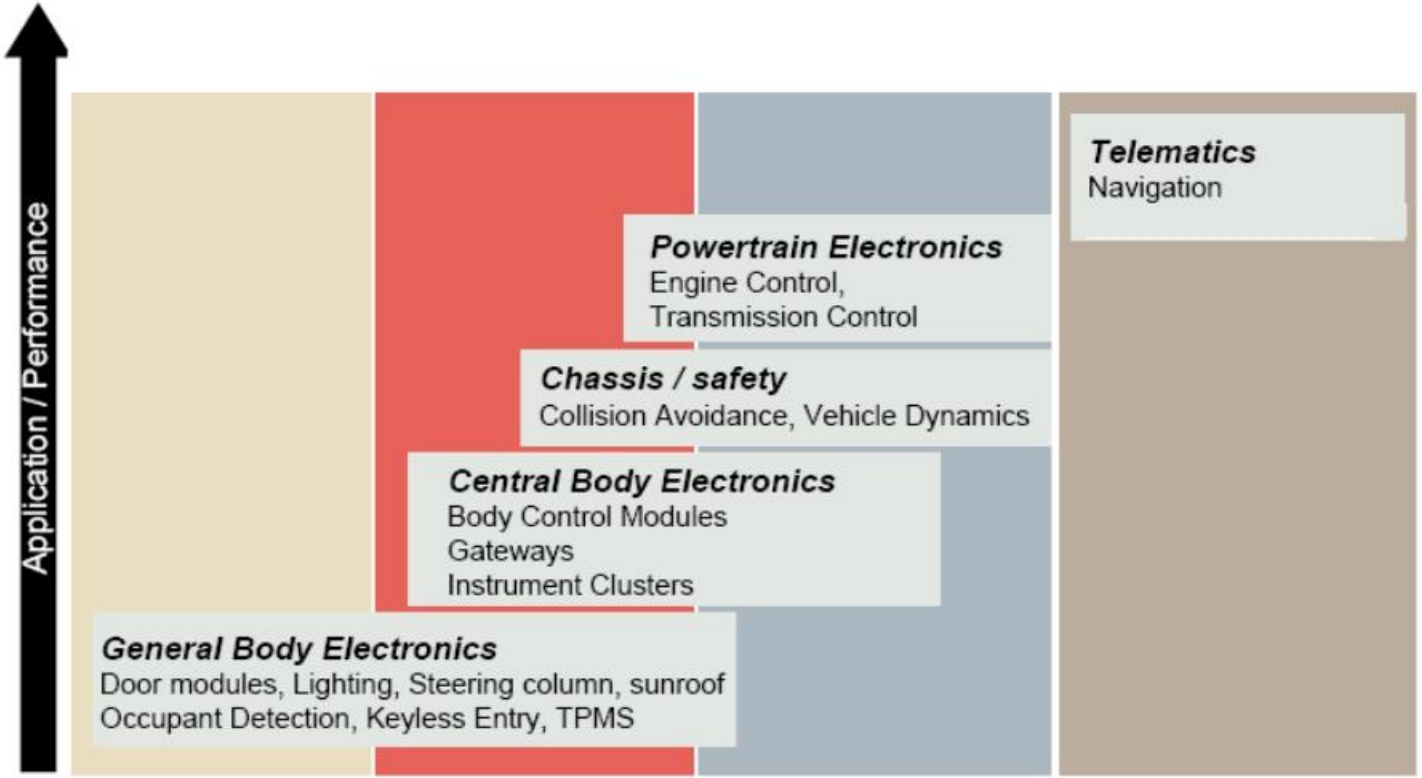
- Transfers data among distributed electronic modules via a serial data bus
- Combine signals on a single wire through multiplexing
- Information sent to individual control modules controlling respective functions
- Makes the wiring connections simple & helps in easy fault detection

Benefits

- Decreased number of dedicated wires
- Improved system cost, weight, reliability, serviceability, and installation.
- Data can be shared, thus eliminating the need for redundant sensors
- Provides greater vehicle content flexibility as functions can be added or deleted with ease



Communication protocol



Where it started

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- Controller Area Network (CAN) is a serial network
- A network among sensors, microcontrollers and actuators
- Two-wire system, well suited for high speed applications using short messages
- Offers communication rate up to 1 Mbits/sec and 64 bytes of data allowing real-time control



Where it started

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- Bus topology -> plug and play
- Bus speed
- Priority of messages
- Multi master
- Bus access control : Non destructive bit wise arbitration
- Small messages (max 8 data bytes) protected by checks



How it evolved

•CAN FD

Increased data size

8 bytes to 64 bytes

Increased data rate

Upto 5Mbit/s

•CAN XL

Increased data size

8 bytes to 2048 bytes

Increased data rate

Upto 10Mbit/s

Why Auto OEMs are shifting from CAN

VEHICLE ARCHITECTURE WITH HIGH PERFORMANCE COMPUTERS

- Bandwidth - ADAS/AD Requirements
- Data Size

Why Auto OEMs are shifting from CAN

CAN, LIN, FlexRay, CAN-FD

Moving to ONE network?
Reuse?

Ethernet in the industry

• The intensive research started more than 10 years ago under the leadership of the **IAONA** (Industrial Automation Open Networking Alliance).

• This initially led to the development of Fast Ethernet by doing this to 100 Mbps. With this, the ground for the desired fast real-time transfer had been prepared.

• There are approximately 20 different protocols for Industrial Ethernet, which are oriented on **IEEE 802.3**.

IEEE 802.3 Ethernet Working Group

The IEEE 802.3 Ethernet Working Group (WG) is responsible for developing the Standard for Ethernet under the auspices of the IEEE 802 LAN/MAN Standards Committee (LMSC).



Ethernet in the industry

•Ethernet has already proven itself as a secure transfer medium that can handle large amounts of data and can reduce weight by 30% over the traditional CAN/LIN harnesses.

•The advantages of Ethernet—multi-point connections, higher bandwidth, and low latency—are attractive to automotive manufacturers

However, traditional Ethernet is too noisy and interference-sensitive to be used in automobiles.



Ethernet in the industry

	Ethernet 100Base-TX
Standard	IEEE 802.3
Data Rate/Base Bandwidth	100 Mbps
Signal	MLT3 – Multi level transmit that uses three voltage levels
Length	100m
Connector	RJ45
Cable	Two twisted pair single direction



Ethernet in the industry

- The auto industry was already beginning to get interested in Ethernet in 2008 onwards
- As with any new technology, though, carmakers were initially tentative
- They picked a few models and built a system model with Ethernet
- The original idea was to use Ethernet to upload software onto on-board diagnostic systems



Why Commercial Ethernet not suited for Automotive Applications?

4 Wires EMC Requirements stringent in Automotive applications

Impedance tolerances and power loss are a big deal in the vehicle environment and they are controlled very tightly

Wakeup Requirements



Evolution of Automotive Ethernet

The Institute of Electrical and Electronic Engineers (IEEE) 802.3bw standard (also known as 100BASE-T1, previously known as BroadR-Reach) is a 100 Mbps automotive Ethernet standard aimed at:

- increasing data throughput
- meeting strong automotive standards and
- reducing cabling weight and cost in automotive networking



Evolution of Automotive Ethernet

The term "Automotive Ethernet" can be used to refer to any Ethernet-based network for in-vehicle electrical systems.

It encompasses 100Base-T1, as well as several other variants and speeds of Automotive Ethernet (e.g., 10Base-T1, 1000Base-T1).

100Base-T1 (100 Mbps Ethernet over single twisted pair which is used for automotive applications)
1000Base-T1 (1000 Mbps ethernet over single twisted pair)
IEEE 802.1 Audio/Video Bridging (AVB) specifies priority queuing, time synchronization and traffic shaping to be used in vehicles.

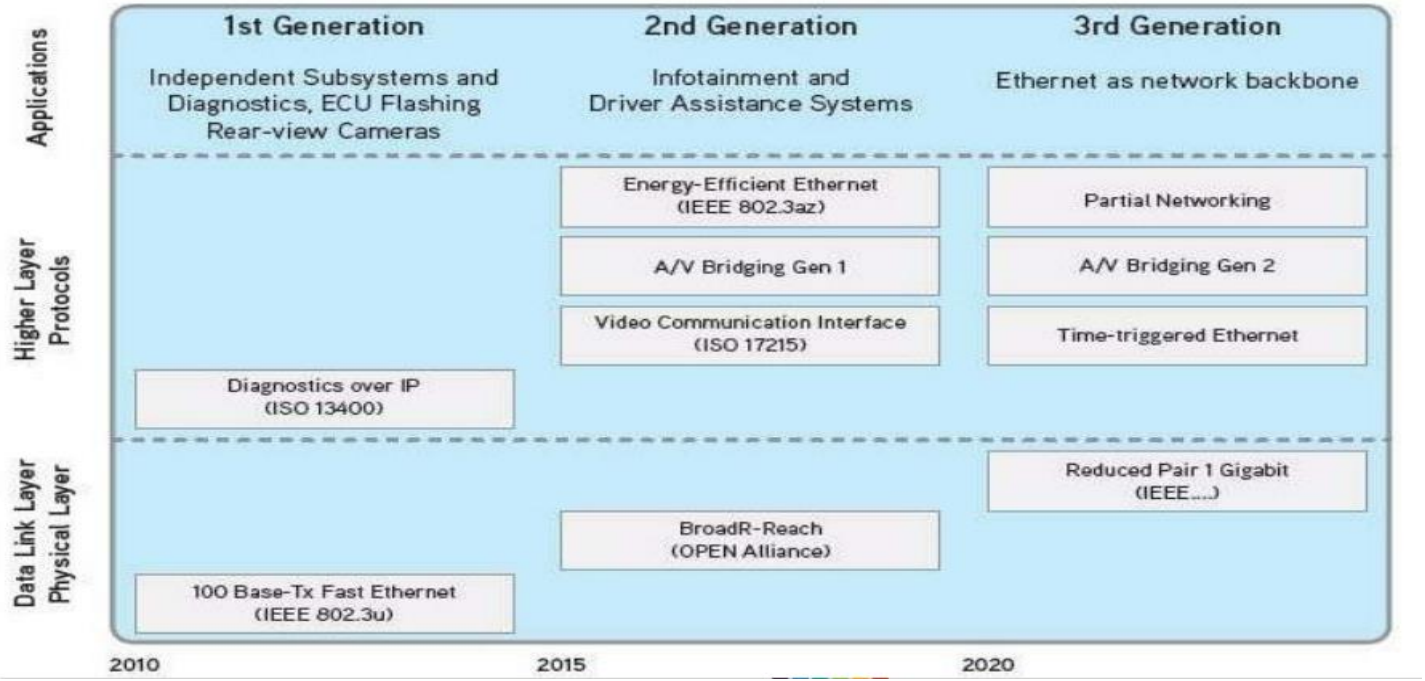


Automotive Ethernet

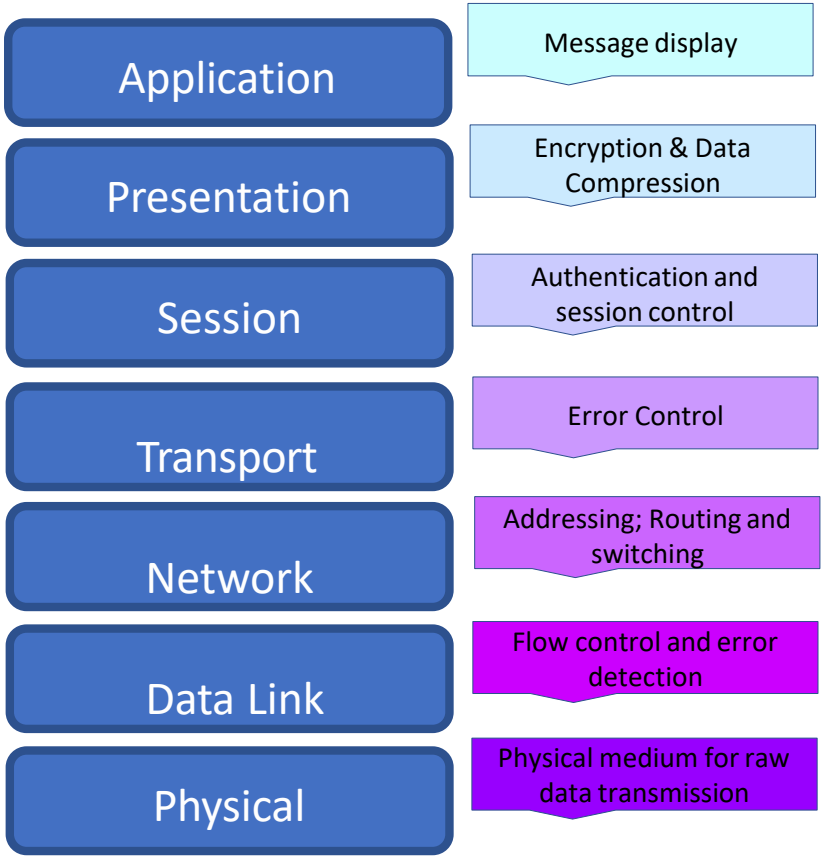
- Supports High Data Rate without EMI/EMC problems
- Meet latency, synchronization and bandwidth requirements
- Power delivered over ethernet cables
- Supports distributed network architecture for communication
- Work with TCP/IP based protocols



Automotive Ethernet

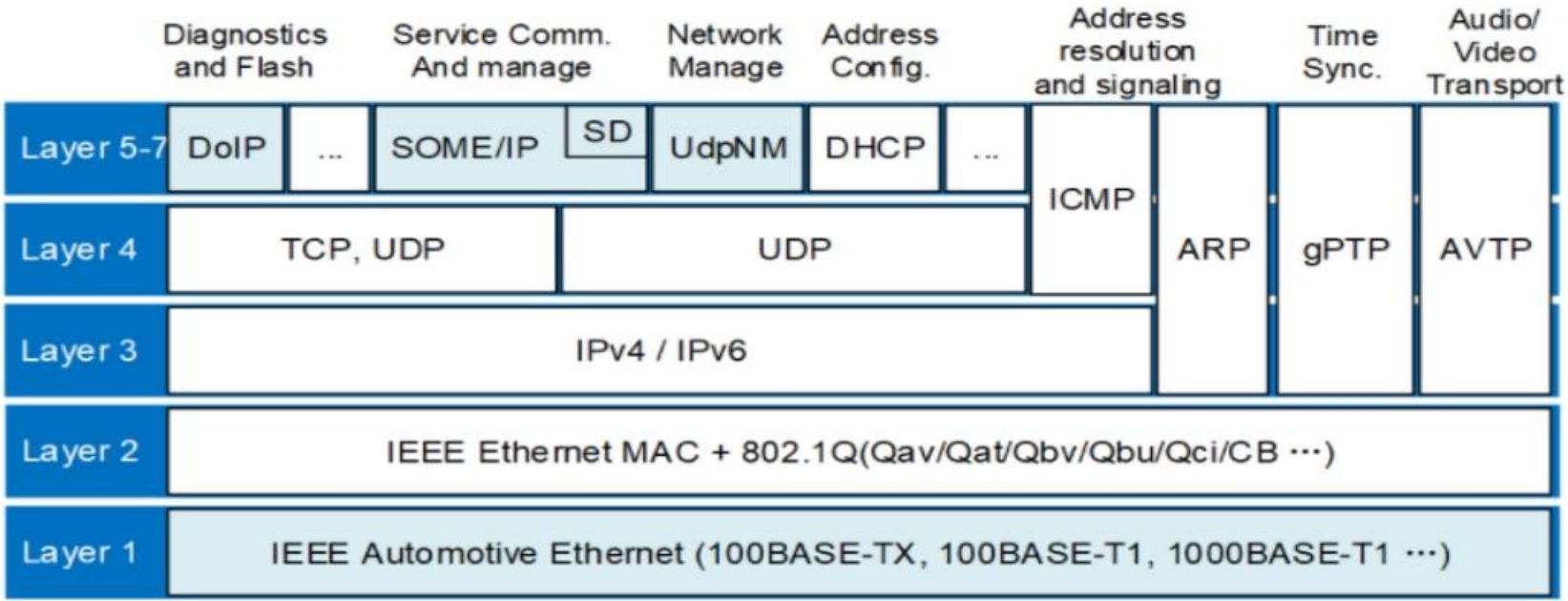


OSI Layers


Application
Message display
Presentation
Encryption & Data Compression
Session
Authentication and session control
Transport
Error Control
Network
Addressing; Routing and switching
Data Link
Flow control and error detection
Physical
Physical medium for raw data transmission

TCP/IP layers Overview

TCP/IP represents a protocol stack, which is composed of a series of small and special protocols including :

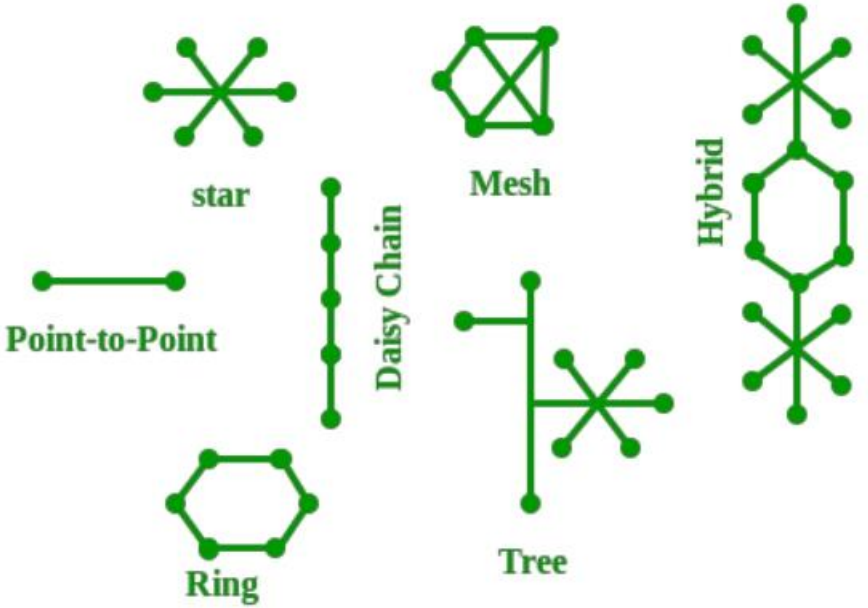


Ethernet in the industry

	Automotive Ethernet 100Base-T1	Automotive Ethernet 1000Base-TX
Standard	IEEE 802.3bw	IEEE 802.3bp
Data Rate/Base Bandwidth	100 Mbps	1000Mbps
Signal	PAM3 – Pulse Amplitude Modulation that uses three voltage levels	PAM3 – Pulse Amplitude Modulation that uses three voltage levels
Length	15m	15m
Connector	Varies by OEMs	Varies by OEMs
Cable	One twisted pair bi-directional	One twisted pair bi-directional

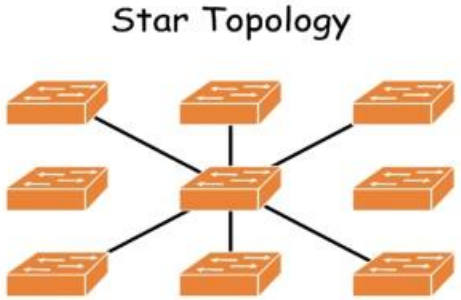
Ethernet networking concepts

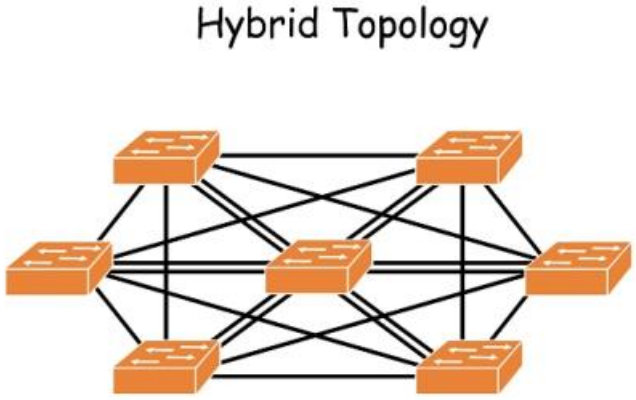
An interconnection of multiple devices, also known as hosts, that are connected using multiple paths for the purpose of sending/receiving data or media.



Ethernet networking concepts

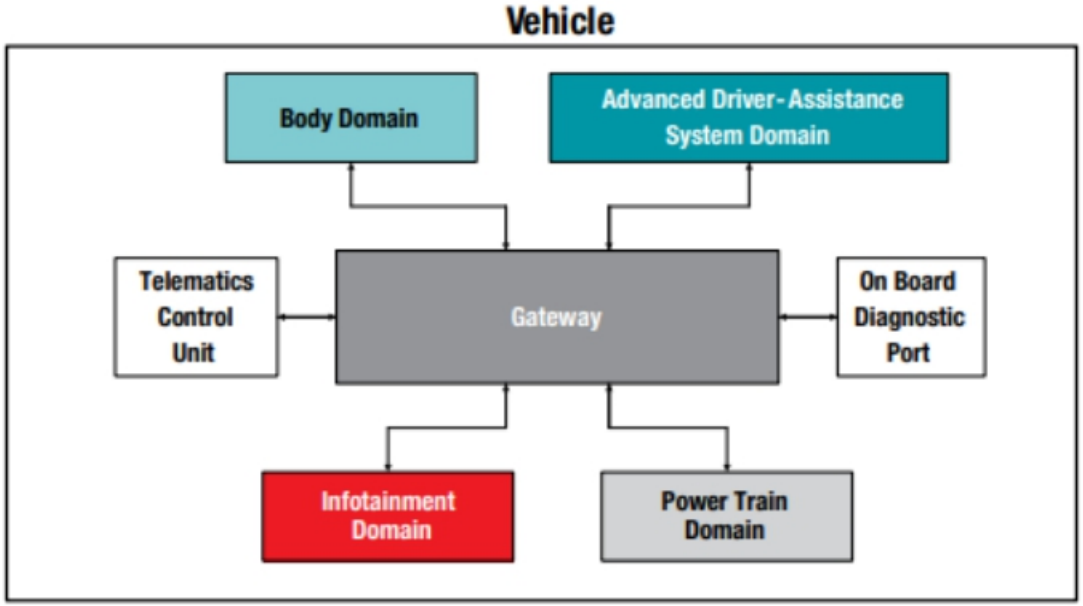
A star topology is just like its name and makes a star formation with a central device that provides all of the connectivity.



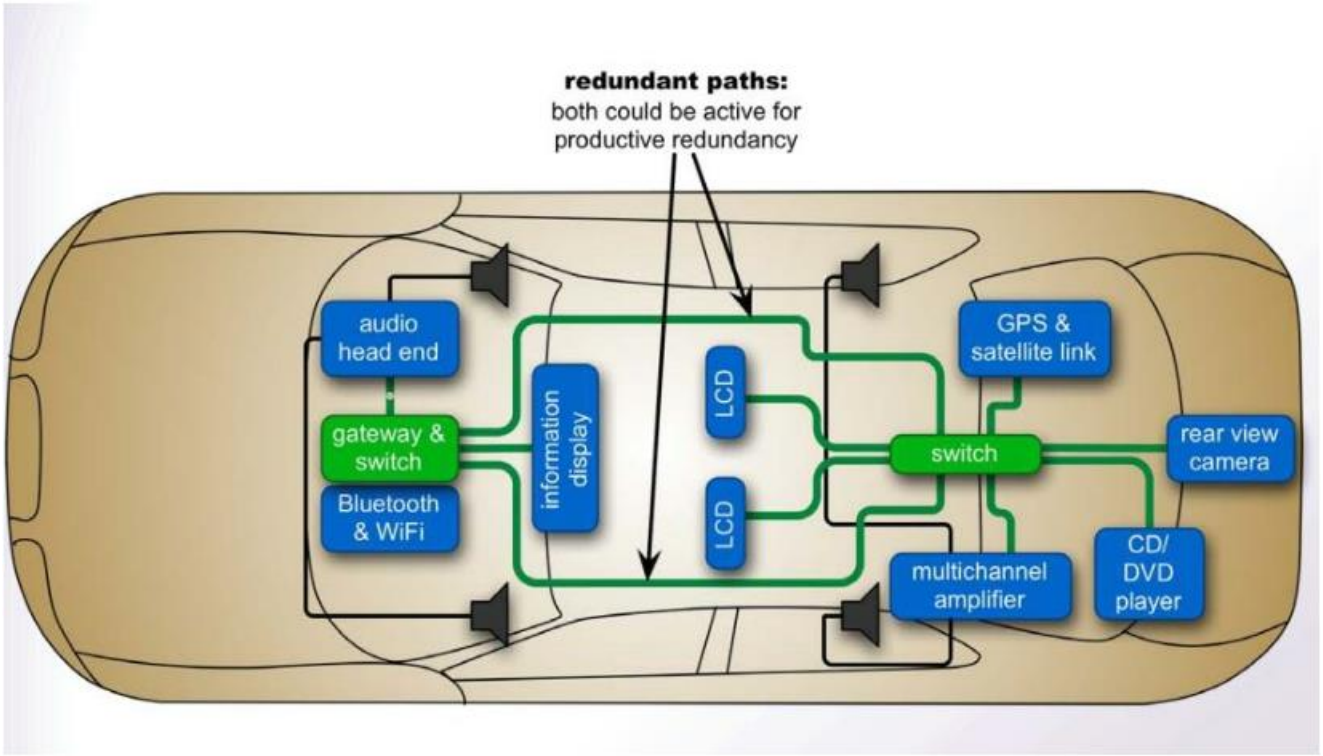


A hybrid topology is created when different types of topologies are combined. For example, if you merged a star and meshed topology than you would have a hybrid topology.

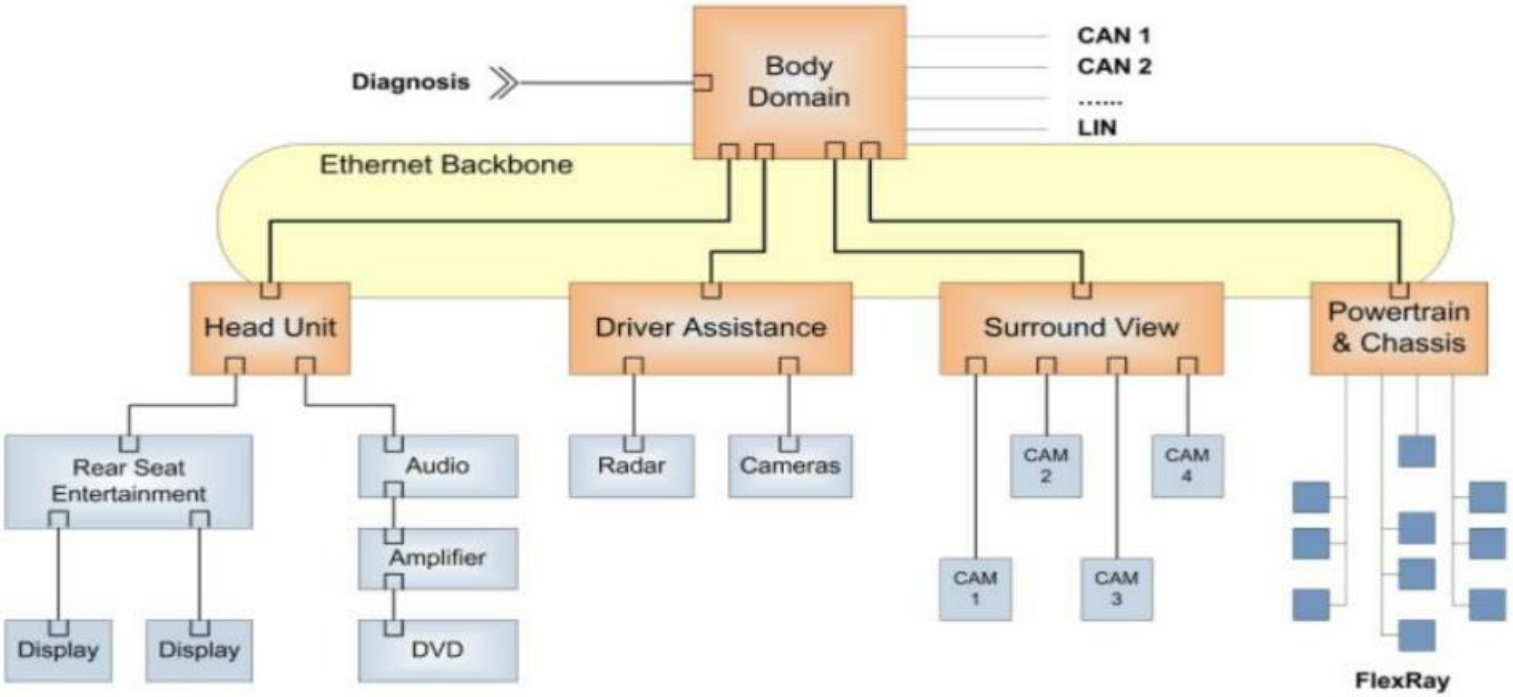
Ethernet networking concepts



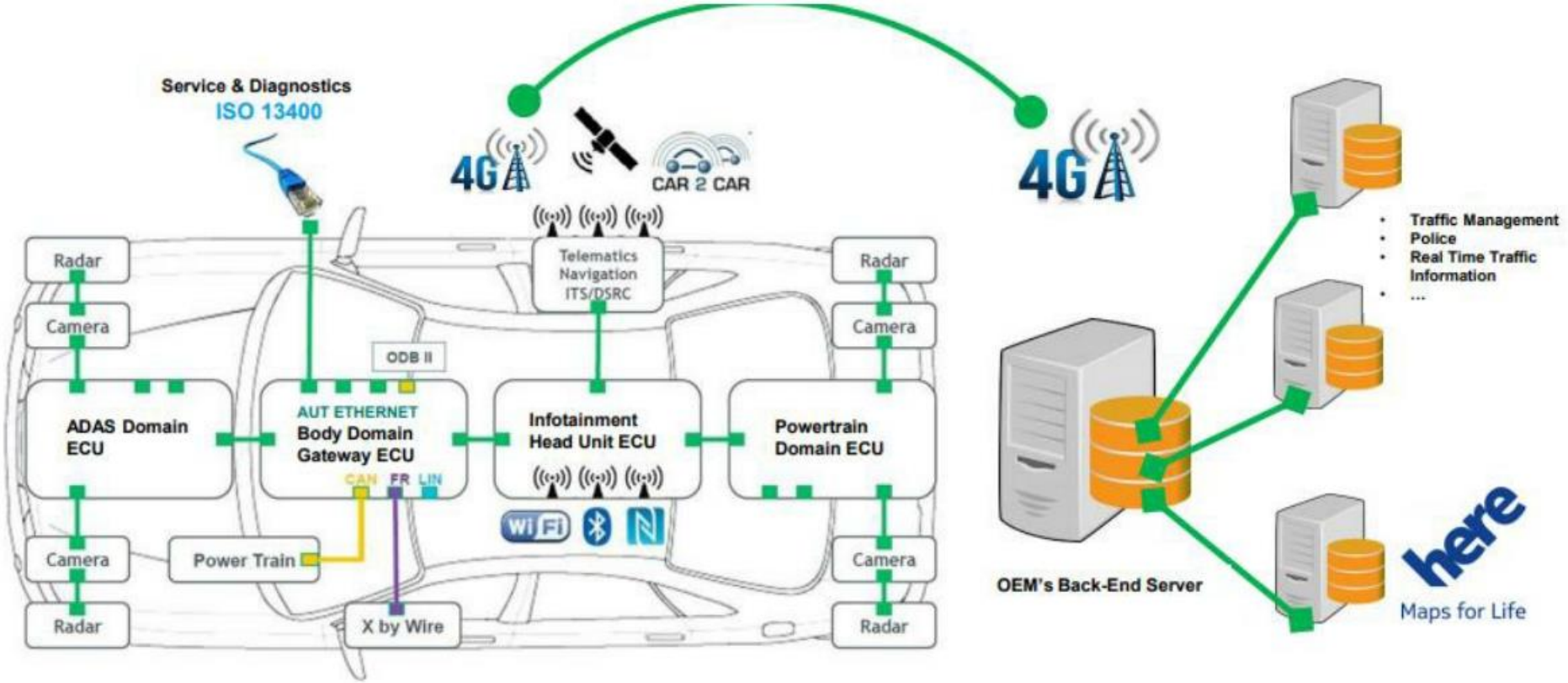
Ethernet networking concepts



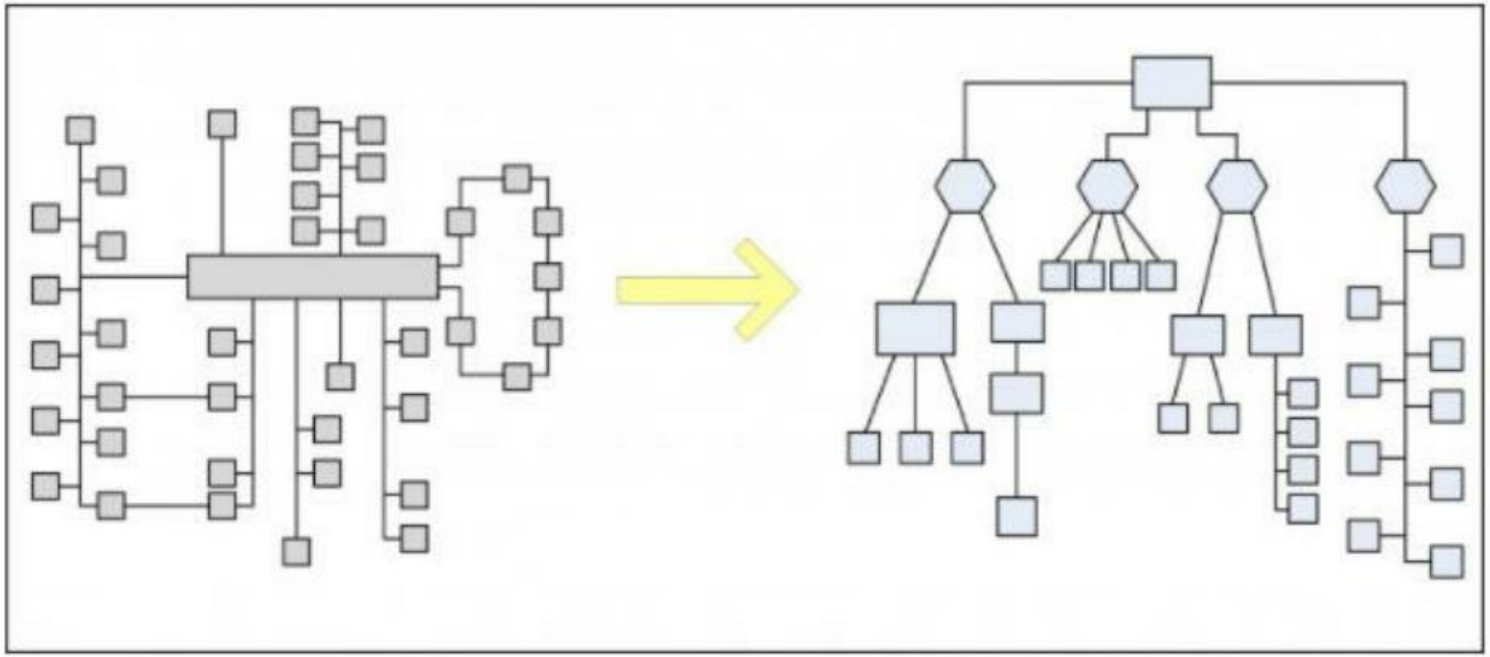
Example :


Diagnosis
Body Domain
CAN 1
CAN 2
.....
LIN
Ethernet Backbone
Head Unit
Driver Assistance
Surround View
Powertrain & Chassis
Rear Seat Entertainment
Audio
Amplifier
DVD
Display
Display
Radar
Cameras
CAM 1
CAM 2
CAM 3
CAM 4
FlexRay

Example- Upcoming Vehicle Architecture :



Example- Upcoming Vehicle Architecture :



Devise Electronics Pvt. Ltd.

HQ: 207-208, Satyam Industrial Estate,
Behind CDSS, Erandwane, Pune,
MS, INDIA - 411004

Ph No: **+91-9552411011**

Email: ***info@deviseelectronics.com***

Web: ***www.deviseelectronics.com***

devise
Electronics Private Limited